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PrimerDesigner: Designing Efficient Primers for Synthesizing Large Protein Libraries Without Cross-Hybridization

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ABSTRACT

Efficient synthesis of large protein libraries is key to making high-throughput biochemistry scalable. Advances in DNA microarray technology now allow more than 170,000 designed oligonucleotides to be synthesized in a single array, dramatically reducing cost and enabling pooled amplification for downstream mutagenesis reactions. Effective amplification of short oligonucleotides into long coding sequences requires efficient primer design (PD) without cross-hybridization. However, no PD method currently guarantees complete coverage of the protein-coding sequences with maximum efficiency and no cross-hybridization over multiple proteins or multiple variants of the same protein. Here, we present PrimerDesigner, a suite of methods to design the most efficient primers for synthesizing large protein libraries with complete protein-coding sequence coverage and without cross-hybridization. We first prove that PD with complete coverage and without cross-hybridization is NP-hard, even for a single protein. As a solution, we formulate the PD problem using integer linear programming (ILP). When no cross-hybridization is possible, we solve the PD problem for a single protein in linear time. Moreover, we extend our ILP formulation to handle multiple variants of the same protein. We demonstrate that PrimerDesigner produces optimal solutions in only a few minutes for a single protein and within a couple of hours for multiple proteins while achieving higher efficiency than a greedy baseline approach. We expect PrimerDesigner to enable the synthesis of large protein libraries at an unprecedented scale and efficiency.

Keywords: cross-hybridization, integer linear programming, NP-hard, primer design.

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1. INTRODUCTION

Recent technological advances in oligonucleotide synthesis and the development of novel high-throughput assays to characterize protein function are enabling a new era of high-throughput biochemistry (Appel et al., 2021). Large pooled libraries of oligonucleotides have been integrated with automated screening platforms to yield high-quality datasets linking amino-acid sequence to functional properties at unprecedented scale. These libraries are used in high-throughput protein screens designed to select variants with desired properties and to explore their fitness landscape. Such approaches significantly accelerate protein engineering and functional genomics research, enabling comprehensive mapping of mutational effects (Fowler and Fields, 2014; Álvarez-Rodríguez et al., 2024; Meiri et al., 2024).

However, protein library construction to date has generally been constrained by the limited oligonucleotide lengths that can be commercially synthesized. With high-throughput biochemistry tools significantly elevating the throughput of quantitative protein measurements by 10^2-10^3 times, the creation of variant libraries has emerged as a critical bottleneck (Fordyce et al., 2010; Tang and Fenton, 2017; Nisthal et al., 2019; Aditham et al., 2021; Markin et al., 2021).

Assays were developed to synthesize large protein libraries while reducing the time and cost of conventional synthesis. These new assays include commercially available oligo arrays, where coding regions are first divided into $\sim$200–300 nucleotide windows. The 5′ and 3′ termini of each window ($\sim$25 nt each) serve as unique primer-binding regions that define a corresponding forward and reverse primer pair for amplification of that window from the pooled array. Consecutive windows overlap by $\sim$20–30 bp to ensure complete coverage of the coding sequence while allowing each window to be independently amplified by its corresponding primer pair. This strategy enables mutagenic oligos for many targets to be encoded by the same parent array, greatly reducing the cost (per oligo) and making it possible to continue to amplify mutagenic oligos from the array via polymerase chain reaction (PCR). The amplified oligos can then be used directly in downstream mutagenesis reactions to generate the desired protein variants (Melnikov et al., 2014; Appel et al., 2021; Álvarez-Rodríguez et al., 2024).

Still, the effectiveness of the PCR reaction depends heavily on the quality of the selected primers. Effective primer design (PD) ensures that each segment of the coding sequence is accurately amplified in high yield without cross-hybridization. Hybridization of primers to nontarget DNA sequences, termed *cross-hybridization*, can lead to nonspecific amplification and errors in the PCR reaction and can ultimately result in a reduction in the yield of correct protein synthesis. This issue is particularly pronounced in mutagenesis libraries containing multiple variants of the same protein or homologous proteins (Aharon et al., 2020; Leander et al., 2022; Meiri et al., 2024). When amplifying such variant libraries from a single oligonucleotide microarray, distinct primer pairs must be designed for each protein variant to enable physical separation through selective amplification of the oligonucleotides belonging to that specific variant.

The objective of the *PD* problem for PCR-based library synthesis is to identify the most efficient primers encompassing all coding sequences without cross-hybridization (Fig. 1A). PD methods were originally developed for microarrays targeting a set of genes (Klau et al., 2007), for assembling a peptide sequence based on mass-spectrometry data (Andreotti et al., 2012), and for targeting viral genomes (Quick et al., 2017; Chen et al., 2026). These methods incorporate filters to assess individual primers based on desirable properties, such as GC content and melting temperature. Some methods integrate set-cover optimization techniques, such as integer linear programming (ILP) and greedy heuristics (Pesole et al., 1998; Wang et al., 2004; Jabado et al., 2006; Bashir et al., 2007; Gardner et al., 2009; Andreotti et al., 2012; Hysom et al., 2012). However, there is currently no publicly available method to find the most efficient primer pairs that cover a complete coding sequence without cross-hybridization for the assembly of multiple proteins or multiple variants of the same protein in the context of PCR-based protein library synthesis (Supplementary Section S1). Existing methods can be used to design primers for individual protein-coding sequences, but they cannot generate a globally optimal PD over multiple proteins while ensuring no intra- or inter-sequence cross-hybridization.

The PD problem for PCR-based protein library synthesis has never been formally defined or analyzed from a computational optimization perspective. Existing methods rely on heuristic or greedy approaches and do not provide a formal problem formulation, computational complexity analysis, or optimality guarantees. To the best of our knowledge, no prior work has formulated the problem as a global optimization problem with complete coverage of coding sequences, cross-hybridization avoidance, and efficiency maximization, nor provided

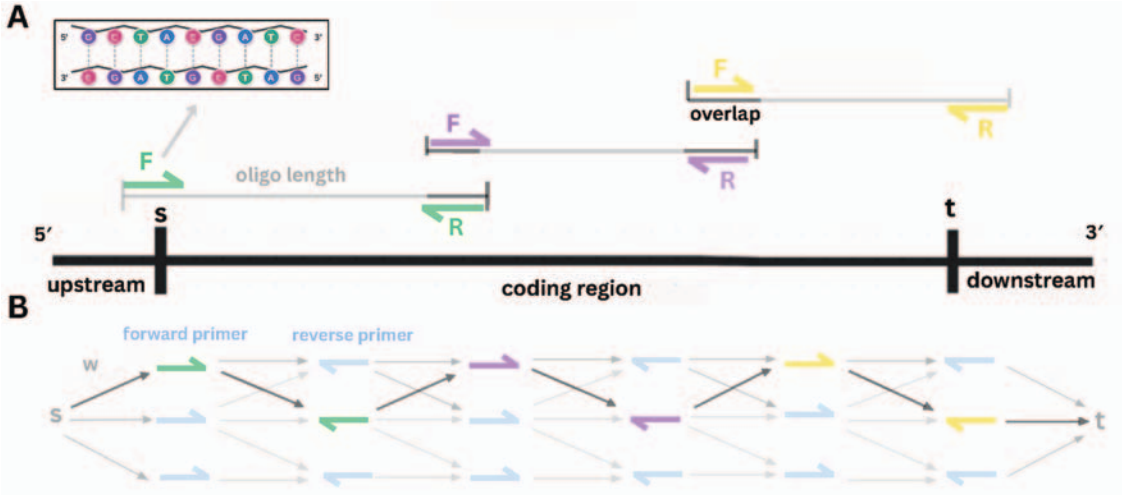


FIG. 1. Primer-design problem and primer graph illustrations. **(A)** Primer-design problem illustration. Given a protein-coding sequence with start and end positions, denoted by s and t, the goal is to find the most efficient forward and reverse primer pairs that cover the whole sequence and do not cross-hybridize. The inset shows an example of a DNA sequence of a primer. **(B)** An illustration of the primer design presented in panel A over a subset of the corresponding primer graph for the DNA sequence. Vertices corresponding to primers are shown in matching colors with their respective primers in panel A. s and t denote the source and target vertices representing the upstream and downstream regions, respectively. The gray arrows represent the edges of the graph, with edge weight w corresponding to the efficiency of the incoming primer. Black arrows indicate the optimal path corresponding to the selected primer design in panel A.

algorithms that guarantee optimal solutions. This limits the ability to design primer sets that are provably optimal and scalable to modern protein synthesis workflows.

Here, we formally define the PD problem as a combinatorial optimization problem of selecting primer pairs that completely cover the protein-coding sequences while avoiding cross-hybridization and maximizing overall primer efficiency. We first prove that the problem is NP-hard in its general form and propose an ILP solution to compute an optimal PD. We then develop additional implementations for the different formulations of the problem. We integrated these solutions into a suite of methods, called PrimerDesigner, which compute optimal primers within minutes for a single protein and within a couple of hours for multiple proteins. We demonstrate through extensive benchmarking that PrimerDesigner outperforms a greedy baseline approach and existing PD methods. We expect PrimerDesigner to significantly advance large-scale protein library synthesis.

2. PRELIMINARIES AND DEFINITIONS

2.1. Preliminaries

Definition 1. *A **DNA sequence** $S \in \Sigma^L$ is an L-long string over $\Sigma = \{A, C, G, T\}$,* that is, $S = s_1, \ldots, s_L$.

Definition 2. *A **subsequence** $S[i,j)$ or $S[i, j]$ is a substring of S from i to j including s_i and excluding or including s_j, respectively.*

Definition 3. *A **primer** p of S is a subsequence of S, and a **forward and reverse** primer pair (f, r) is a pair of nonoverlapping subsequences of S, where f precedes r.*

Definition 4. *The hybridization **efficiency** of a primer p, denoted by $E_{ff}(p)$, is a numerical score representing the expected hybridization performance of p. The score depends on the primer's melting temperature, GC content, self-complementarity, number of adenines at the 3′ terminus, and thermodynamic stability at the 3′ terminus. Details of its computation are provided in Section 3.5.*

Definition 5. *A **forbidden** primer pair (p, q) is a pair of primers that cannot be simultaneously selected due to their **cross-hybridization risk**, which is the risk of a primer hybridizing to a DNA sequence other than its intended target. A primer pair (p, q) is considered to have cross-hybridization risk if the predicted heterodimer melting temperature between p and q exceeds 45°C.*

2.2. Problem definitions

We consider three related PD problems (Fig. 2). Maximum-efficiency primer design covering a single protein (MEPD-CSP) addresses the design of primers covering a single protein-coding sequence (Fig. 2A), and MEPD-CSP-r is a relaxed version that assumes there are no forbidden primer pairs. Maximum-efficiency primer design covering multiple proteins (MEPD-CMP) extends this formulation to multiple protein-coding sequences while avoiding cross-hybridization among selected primers (Fig. 2B). Finally, maximum-efficiency primer design covering multiple protein variants (MEPD-VAR) considers aligned protein variants and prevents the selection of primers with high overlap across variants (Fig. 2C).

2.2.1. Maximum-efficiency primer design covering a single protein (MEPD-CSP).

- **Instance:**
  - A DNA sequence S, s, and t start and end positions in the sequence, minimum and maximum primer length $p_{min/max}$, minimum and maximum oligo length $o_{min/max}$, minimum and maximum oligo overlap $v_{min/max}$, primer hybridization efficiency function E_{ff}, and a set of forbidden primer pairs F.

- **Valid solution:**
  - A choice of $n_p \geq 1$ ordered forward and reverse primer pairs $P = \{(f_\ell = S[i^f_\ell, j^f_\ell),\ r_\ell = S[i^r_\ell, j^r_\ell))|1 \leq \ell \leq n_p\}$ such that $p_{min} \leq j^f_\ell - i^f_\ell, j^r_\ell - i^r_\ell \leq p_{max}$, $o_{min} \leq j^r_\ell - i^f_\ell \leq o_{max}$, $v_{min} \leq j^r_\ell - i^f_{\ell+1} \leq v_{max}$ for $1 \leq \ell < n_p$, $i^f_1 \leq s$ and $j^r_{n_p} \geq t$, and no pair of any two primers is in F.

- **Goal:**
  - Maximize $\sum_{\ell=1}^{n_p} \left(E_{ff}(f_\ell) + E_{ff}(r_\ell)\right)$.

We define MEPD-CSP-r when $F = \emptyset$, which is motivated by the fact that the cross-hybridization risk between primers of the same coding-sequence is unlikely (Supplementary Section S2).

2.2.2. Maximum-efficiency primer design covering multiple proteins (MEPD-CMP).

- **Instance:**
  - M DNA sequences $S = \{S_m|1 \leq m \leq M\}$, s_m and t_m start and end positions of sequence S_m, minimum and maximum primer length $p_{min/max}$, minimum and maximum oligo length $o_{min/max}$, minimum and maximum oligo overlap $v_{min/max}$, primer hybridization efficiency function E_{ff}, and a set of forbidden primer pairs F.

- **Valid solution:**
  - A choice of $n_{m,p} \geq 1$ ordered forward and reverse primer pairs for each sequence S_m: $P_m = \{(f_{m,\ell} = S_m[i^f_{m,\ell}, j^f_{m,\ell}), r_{m,\ell} = S_m[i^r_{m,\ell}, j^r_{m,\ell}))|1 \leq \ell \leq n_{m,p}\}$ such that $p_{\min} \leq j^f_{m,\ell} - i^f_{m,\ell}, j^r_{m,\ell} - i^r_{m,\ell} \leq$

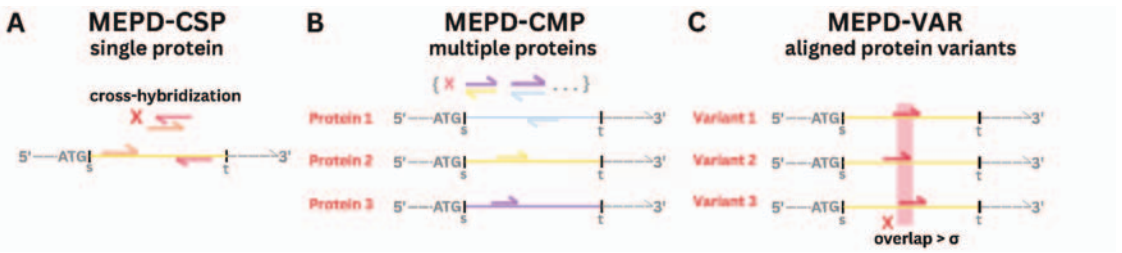


FIG. 2. Overview of the primer design problems considered in this study. **(A)** MEPD-CSP designs primers covering a single protein-coding sequence. **(B)** MEPD-CMP extends the problem to multiple protein-coding sequences while avoiding cross-hybridization among selected primers. **(C)** MEPD-VAR considers aligned protein variants, where primer pairs with overlap greater than σ are forbidden. Colored directional segments denote selected primer pairs and red X symbols indicate forbidden primer combinations. MEPD-CMP, maximum-efficiency primer design covering multiple proteins; MEPD-CSP, maximum-efficiency primer design covering a single protein; MEPD-VAR, maximum-efficiency primer design covering multiple protein variants.

p_{max}, $o_{min} \le j^r_{m,\ell} - i^f_{m,\ell} \le o_{max}$, $v_{min} \le j^r_{m,\ell} - i^f_{m,\ell+1} \le v_{max}$ for $1 \le \ell < n_{m,p}$, $i^f_{m,1} \le s_m$ and $j^r_{m,n_{m,p}} \ge t_m$, and no pair of selected primers from $\cup_{m=1}^{M} P_m$ is in F.

- **Goal:**
  - Maximize $\sum_{m=1}^{M} \sum_{\ell=1}^{n_{m,p}} (E_{ff}(f_{m,\ell}) + E_{ff}(r_{m,\ell}))$.

2.2.3. Maximum-efficiency primer design covering multiple protein variants (MEPD-VAR). We define MEPD-VAR when F is the set of all pairs of primers that are not from the same protein and overlap in sequence indices by more than σ nucleotides, which is a proxy for cross-hybridization risk since high sequence similarity leads to high predicted melting temperatures. This formulation is motivated by the need to generate libraries of highly similar proteins or even variants of the same protein. If the protein sequences are not aligned, they can be aligned to a common coordinate system (e.g., via a multiple sequence alignment), so indices are consistent across proteins. Let M be the number of protein-coding DNA sequences, and let S_m denote the mth sequence. For each $1 \le m \le M$, define the set of valid primer index pairs

$$\mathcal{I}_m = \{(i,j) | 1 \le i \le j \le |S_m|,\ p_{\min} \le j - i + 1 \le p_{\max}\},$$

and let $p^m_{i,j}$ denote a primer on S_m spanning $(i,j) \in \mathcal{I}_m$.

The forbidden primer-pair set is

$$F = \left\{ \left(p^{m_1}_{i_1,j_1}, p^{m_2}_{i_2,j_2}\right) \;\middle|\; \begin{array}{l} 1 \le m_1, m_2 \le M,\ m_1 \ne m_2, \\ (i_1,j_1) \in \mathcal{I}_{m_1},\ (i_2,j_2) \in \mathcal{I}_{m_2}, \\ \text{overlap}\left(p^{m_1}_{i_1,j_1}, p^{m_2}_{i_2,j_2}\right) > \sigma \end{array} \right\}.$$

where

$$\text{overlap}\left(p^{m_1}_{i_1,j_1}, p^{m_2}_{i_2,j_2}\right) = \max\Big(0,\ \min(j_1,j_2) - \max(i_1,i_2) + 1\Big).$$

We prove that MEPD-CSP and MEPD-CMP are NP-hard by a reduction from 3SAT (Supplementary Section S3).

3. METHODS

3.1. Primer graph transformation

We transform the problem of finding a set of optimal forward–reverse primer pairs into a problem of finding a longest path in a primer graph. We define a *primer graph* as a directed acyclic graph that contains source and target vertices s and t, respectively, and represents all possible valid sets of primer choices for a given sequence (Fig. 1B). Each vertex in the primer graph represents a unique forward or reverse primer. The outgoing neighbors of a forward-primer vertex are all possible reverse primers that can be paired with this primer given oligo- and primer-length constraints. The outgoing neighbors of a reverse-primer vertex are all possible forward primers that can be part of the next primer pair given overlap and primer-length constraints. The weight of an edge (v_i, v_j) is $E_{ff}(v_j)$, the efficiency of its target.

3.2. Integer linear programming for MEPD-CSP

We define a binary variable $x_{i,j}$ for every edge (v_i, v_j) in the primer graph $G = (V, E)$ to indicate whether the single-path solution contains the edge. The single-path constraint ensures that only one path, which starts at s, ends at t, and does not start or end at any other vertex, is selected. The forbidden-pair constraints ensure that forbidden primer pairs are not included in the solution. The objective function aims to maximize the efficiency of the path:

$$\text{maximize} \sum_{(v_i,v_j)\in E} E_{ff}(v_j) \cdot x_{i,j}$$

Constraints

1. Single-path constraints:

$$\sum_{(v_i,v_j)\in E} x_{i,j} - \sum_{(v_k,v_i)\in E} x_{k,i} = \begin{cases} 1 & v_i = s \\ -1 & v_i = t \\ 0 & v_i \notin \{s,t\} \end{cases}, \quad \forall v_i \in V$$

2. Forbidden-pair constraints:

$$\sum_{(v_i,v_j)\in E} x_{i,j} + \sum_{(v_\ell,v_k)\in E} x_{\ell,k} \leq 1, \quad \forall (v_i, v_\ell) \in F$$

3.3. Integer linear programming for MEPD-CMP

Similarly, we present an ILP formulation for MEPD-CMP. Given M proteins, we create a primer graph $G_m = (V_m, E_m)$ for each protein $1 \leq m \leq M$. Then, the objective function is:

$$\text{maximize} \sum_{m=1}^{M} \sum_{(v_i,v_j)\in E_m} E_{ff}(v_j) \cdot x_{i,j}$$

Constraints

1. Single-path constraints per graph G_m:" where G_m matches the notation used in the sentence above.

$$\sum_{(v_i,v_j)\in E_m} x_{i,j} - \sum_{(v_k,v_i)\in E_m} x_{k,i} = \begin{cases} 1 & v_i = s_m \\ -1 & v_i = t_m, \quad \forall v_i \in V_m, \\ 0 & \text{else} \qquad 1 \leq m \leq M \end{cases}$$

2. Forbidden-pair constraints:

$$\sum_{m=1}^{M} \sum_{(v_i,v_j)\in E_m} x_{i,j} + \sum_{m=1}^{M} \sum_{(v_\ell,v_k)\in E_m} x_{\ell,k} \leq 1, \quad \forall (v_i, v_\ell) \in F$$

3.4. Integer linear programming for MEPD-VAR

We use a similar ILP formulation as MEPD-CSP but modify the forbidden-pair and single-path constraints. Here, σ denotes the overlap limit allowed between primers beyond which cross-hybridization may occur, and M denotes the number of variants. The sequence S represents the consensus coordinate system of all aligned coding sequences, and all constraints below are applied to the primer graph $G = (V, E)$ constructed over S. We define the set $B^f_{i,j}$ (and $B^r_{i,j}$) as the set of forward (and reverse) primers containing $S[i, j]$, that is, $B^f_{i,j} = \{f_\ell | i^f_\ell \leq i < j \leq j^f_\ell\}$. The modified forbidden-pair constraints allow at most one primer to be selected from each set:

$$\sum_{v_\ell \in B^a_{i,i+w}} \sum_{(v_\ell,v_k)\in E} x_{\ell,k} \leq 1, \qquad 0 \leq i \leq |S| - w,\ \sigma + 1 \leq w \leq p_{\max},\ a \in \{f, r\}.$$

The modified single-path constraint ensures that the solution contains exactly M paths that all start at s and end at t:

$$\sum_{(v_i,v_j)\in E} x_{i,j} - \sum_{(v_k,v_i)\in E} x_{k,i} = \begin{cases} M & v_i = s \\ -M & v_i = t \\ 0 & v_i \notin \{s,t\} \end{cases} \quad \forall v_i \in V$$

3.5. Primer efficiency and cross-hybridization risk: Definition and computation

We adopt the primer efficiency formulation by Zeng et al. (2022), which models primer quality as a function of several sequence features. The efficiency for each primer is calculated based on its melting temperature (*tm*), GC content, self-complementarity, the number of adenines at the 3′ terminus, and its thermodynamic stability. For each of these features, a piecewise logistic function is used to assign a normalized score. The overall primer efficiency is defined as the mean of the five feature scores (Supplementary Section S4).

To identify all primer pairs with cross-hybridization risk, that is, set F, we implement a brute-force search that computes the *tm* between every possible primer pair, for both intra-protein and inter-protein cross-hybridizations, using the `calc_heterodimer` function of the `primer3-py` Python package (Koressaar and Remm, 2007). The reported melting temperature *tm* is derived from Primer3′s nearest-neighbor thermodynamic model, which accounts for the Gibbs free energy of the heterodimer interaction. The asymptotic runtime of this brute-force search for inter-protein cross-hybridization is

$$T_{\text{inter}} = \sum_{\ell_1=p_{\min}}^{p_{\max}} \sum_{\ell_2=p_{\min}}^{p_{\max}} \sum_{i=1}^{L_1-\ell_1+1} \sum_{j=1}^{L_2-\ell_2+1} T(\text{calc_heterodimer}),$$

and for intra-protein cross-hybridization

$$T_{\text{intra}} = \sum_{\ell_1=p_{\min}}^{p_{\max}} \sum_{\ell_2=p_{\min}}^{p_{\max}} \sum_{i=1}^{L-\ell_1+1} \sum_{j=i+\ell_1}^{L-\ell_2+1} T(\text{calc_heterodimer}).$$

where L_1, L_2 denote the two inter-protein sequence lengths and L denotes the intra-protein sequence length.

Each call to `calc_heterodimer` between two primers of lengths ℓ_1 and ℓ_2 takes $T(\text{calc_heterodimer}) = O(\ell_1\ell_2)$ since `calc_heterodimer` performs pairwise alignment by dynamic programming.

3.6. PrimerDesigner implementation details

We integrated our solutions for the different formulations of the PD problem into a unified framework of methods called *PrimerDesigner*, which provides dedicated implementations for each formulation. The different implementations and their corresponding problem formulations are summarized in Table 1.

PD-single-ILP solves the PD problem for a single protein by formulating it as an ILP with forbidden-pair constraints on all cross-hybridizing primer pairs within the same protein.

TABLE 1. PRIMERDESIGNER IMPLEMENTATIONS: PROBLEM FORMULATIONS, ALGORITHMS, OPTIMALITY, AND WORST-CASE RUNTIMES

Implementation	*Problem*	*Algorithm*	*Optimal*	*Worst-case runtime*
PD-single-ILP	MEPD-CSP	ILP	Yes[a]	Exponential time
PD-single-LPath	MEPD-CSP-r	Longest path	Yes	$O(\|V\|+\|E\|)$
PD-mul-Greedy	MEPD-CMP	Greedy	No	$O\left(\sum_{m=1}^{M}(\|V_m\|\|E_m\|+\|V_m\|^2)\right)$
PD-mul-ILP	MEPD-CMP	ILP	Yes[a]	Exponential time
PD-var-ILP	MEPD-VAR	ILP	Yes[a]	Exponential time
PD-var-Greedy	MEPD-VAR	Greedy	No	$O\left(M(\|V\|+\|E\|)\right)$

[a]For ILP-based methods, optimality is guaranteed only when the solver terminates with MIP gap = 0.

ILP, integer linear programming; MEPD-CMP, maximum-efficiency primer design covering multiple proteins; MEPD-CSP, maximum-efficiency primer design covering a single protein; MEPD-VAR, maximum-efficiency primer design covering multiple protein variants; PD, primer design.

PD-single-LPath solves the PD problem for a single protein under the assumption that there is no cross-hybridization risk within the same protein (Supplementary Section S2), by finding the longest path from s to t in the primer graph.

PD-mul-ILP finds optimal primers for multiple nonhomologous proteins. It identifies all cross-hybridization risks using a brute-force algorithm and formulates the PD problem as an ILP with forbidden-pair constraints on all cross-hybridizing primer pairs across the protein-coding sequences.

PD-mul-Greedy applies a greedy strategy to design primers for multiple nonhomologous proteins. At each iteration, primers for the next protein are found using the longest-path algorithm on that protein's graph. If any of these primers cross-hybridize with primers selected for earlier proteins, the conflicting candidate primers are removed from the graph, and the process is repeated. The complete pseudocode is provided in Supplementary Algorithm S1.

PD-var-ILP uses an ILP formulation to design primers for multiple variants of the same protein by adding forbidden-pair constraints that prevent the selection of primers that overlap beyond a predefined threshold.

To prevent the selection of primers with identical subsequences that are nonoverlapping, we add an optional feature to PD-var-ILP that merges sets corresponding to identical subsequences.

PD-var-Greedy applies a greedy approach to choose primers for multiple variants of the same protein. The algorithm computes the longest path for one variant, removes from the primer graph all primers that overlap with any primer in the selected path beyond the predefined overlap threshold, and repeats this process until all variants are covered.

MEPD-CSP and MEPD-CMP are NP-hard, and their ILP formulations may require exponential runtime in the worst case. PD-single-LPath solves the relaxed problem MEPD-CSP-r, which reduces to finding a longest path in a directed acyclic graph and is therefore solvable optimally in linear time. The runtime bounds for the graph-based algorithms are expressed in terms of the size of the primer graph $G = (V, E)$. For a single protein with sequence length $|S|$, the number of primer vertices is bounded by $|V| \leq 2\sum_{\ell=p_{\min}}^{p_{\max}} (|S| - \ell + 1)$ (since in practice vertices with in- and out-degrees zero are removed). To bound $|E|$, let $\Delta_p = p_{\max} - p_{\min} + 1$, $\Delta_o = o_{\max} - o_{\min} + 1$, and $\Delta_v = v_{\max} - v_{\min} + 1$. A forward-primer vertex has at most $\Delta_o \Delta_p$ outgoing edges, while a reverse-primer vertex has at most $\Delta_v \Delta_p$, giving $|E| \leq \frac{|V|}{2} \Delta_p (\Delta_o + \Delta_v) = O\left(|V| \Delta_p (\Delta_o + \Delta_v)\right)$. For multiple proteins indexed by $1 \leq m \leq M$, each with its own graph G_m, $|V_m|$ and $|E_m|$ satisfy the same bounds, with $|S|$ replaced by $|S_m|$, the length of protein m.

PrimerDesigner can also filter inefficient primer candidates by user-defined thresholds. The efficiency thresholds include the maximum difference in *tm* between the forward and reverse primers in each pair, individual-primer *tm*, and GC content.

3.7. Baseline methods for comparison

We compared PrimerDesigner against existing PD tools, published experimental PDs, and baseline approaches appropriate for each problem formulation.

For the single-protein setting, we compared PD-single-LPath against PrimalScheme (Kent et al., 2024) and Olivar (Wang et al., 2024), which are state-of-the-art tools for tiled amplicon PD that generate tiled primer sets and produce outputs comparable to those of PrimerDesigner. In addition, we compared PD-single-LPath against manual PDs reported in experimental studies by Appel et al. (2021) and Álvarez-Rodríguez et al. (2024), which used similar protein library synthesis strategies, to assess how PrimerDesigner performs relative to experimentally validated PDs used in practice.

For the multiple nonhomologous protein setting, we compared our PD-mul-ILP implementation against PD-mul-Greedy, as we are not aware of any existing methods for this problem formulation.

For the multi-variant setting, we compared PD-var-ILP against PD-var-Greedy, as no comparable methods have been proposed to the best of our knowledge.

4. RESULTS

4.1. PrimerDesigner results for a single protein

We first compared the design generated by PD-single-LPath (Supplementary Section S5) to those produced by PrimalScheme (Kent et al., 2024) and Olivar (Wang et al., 2024), as well as to the experimentally validated

design reported by Appel et al. (2021). The comparison was performed on the 1626-nt coding sequence of the SpAP gene, along with its 87-nt upstream and 193-nt downstream segments, which was the focus of the Appel et al. study.

We set the parameters accordingly: $v_{\min}, v_{\max} = 45, 50; o_{\min}, o_{\max} = 195, 205; p_{\min}, p_{\max} = 18, 30$, and evaluated primer efficiency. We measured the runtime and maximum memory usage of PD-single-LPath, PrimalScheme, and Olivar, excluding the method of Appel et al. since it relied on a manual design procedure. We also evaluated whether these improvements translated into higher PCR efficiency (Supplementary Section S6). We performed all runs on a Linux server with 2 Intel(R) Xeon(R) Gold 6338 CPUs @ 2.00 GHz and 512 GB of RAM. PD-single-LPath achieved a mean primer efficiency of 0.49 with 12 primer pairs in 90 seconds, using 2.7 GB of peak memory. PrimalScheme produced a design with a mean primer efficiency of 0.27 across 11 primer pairs in 3 seconds, at a peak memory usage of 59 MB. Olivar generated 15 primer pairs with a mean primer efficiency of 0.20 in 63 seconds, at a peak memory usage of 155 MB. The experimentally validated PD by Appel et al. had a lower mean primer efficiency of 0.17 (13 primer pairs), suggesting that PD-single-LPath may improve PCR amplification efficiency.

We next compared a PD generated by PD-single-LPath to the design of Álvarez-Rodríguez et al. (2024) on a segment of the Coxsackievirus B3 genome. The same parameters were used as above, except for $o_{\min}$ and $o_{\max}$, which were adjusted to 240 and 260, respectively, to match the $\sim$250 nt oligonucleotide length used in their study. PD-single-LPath achieved a mean primer efficiency of 0.46 using 22 primer pairs in 6.4 minutes with a peak memory usage of 12.2 GB, whereas the method of Álvarez-Rodríguez et al. yielded a mean primer efficiency of 0.26 with 18 primer pairs. A comparison of runtime and memory usage with the method of Álvarez-Rodríguez et al. was not possible, as their study did not report a computational PD workflow but rather relied on manually designed primers.

4.2. *PrimerDesigner results for multiple nonhomologous proteins*

We evaluated the performance of PD-mul-ILP on sets of 2–10 nonhomologous proteins. We retrieved the protein-coding sequences from the CCDS database (Pujar et al., 2018) (ordered in Supplementary Table S1). We set all parameter values as in the single-protein case, and varied the number of proteins from 2 to 10, while using the same 87-nt upstream and 193-nt downstream sequences for each protein as in Appel et al. (2021). We compared PD-mul-ILP with a greedy baseline, PD-mul-Greedy, and evaluated the runtime, memory usage, and solution efficiency of both methods. We used the same Linux server as in the single-protein case and Gurobi for solving the ILP (Gurobi Optimization, LLC, 2023).

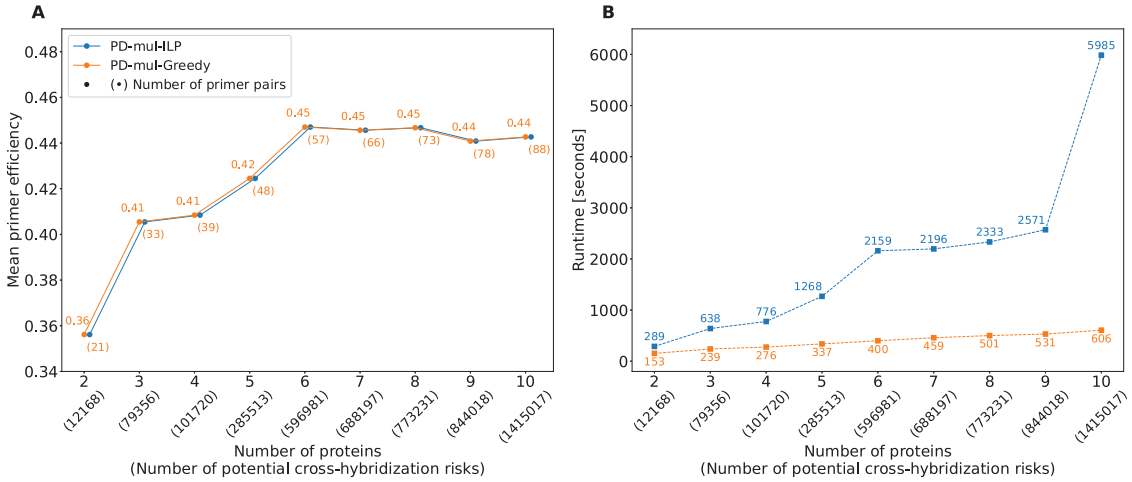


FIG. 3. Performance of PD-mul-ILP compared with PD-mul-Greedy on multiple nonhomologous proteins. We gauged the **(A)** mean primer efficiency and number of primer pairs, and **(B)** runtime. Protein identifiers are in Supplementary Table S1. ILP, integer linear programming; PD, primer design.

The mean primer efficiency and the selected primer pairs obtained by PD-mul-ILP were identical to those obtained by PD-mul-Greedy across all tested protein subsets, and PD-mul-ILP completed in under 2 hours (Fig. 3A, B). The PD-mul-Greedy approach was much faster ($p = 0.002$, Wilcoxon signed-rank test), but these runtimes are negligible compared to the time required to conduct the experiments. Peak memory usage remained low and nearly constant for PD-mul-Greedy, whereas PD-mul-ILP required substantially more memory as the number of proteins increased (Supplementary Fig. S4A).

During the PD-mul-Greedy runs, no reiterations occurred as a result of cross-hybridizations with primers selected for earlier proteins. We speculate that, due to the negligible cross-hybridization risks between nonhomologous proteins and within a protein (Supplementary Section S2), PD-mul-Greedy performed as well as PD-mul-ILP. Since PD-mul-Greedy produced identical PDs to PD-mul-ILP while requiring substantially less runtime and memory, we recommend using this approach for multiple nonhomologous proteins.

We applied PD-mul-Greedy to 100 protein-coding sequences retrieved from the CCDS database (Supplementary Table S2, available in our GitHub repository). It ran in less than 80 minutes with a peak memory usage of 2.22 GB and found 600 primer pairs in total that fully covered the proteins without any cross-hybridization. During the run, there were 135 reiterations due to discovered cross-hybridizations with primers selected for earlier proteins. These reiterations occurred in 15 of the 100 proteins.

4.3. PrimerDesigner results for multiple variants of the same protein

We evaluated the performance of PD-var-ILP on multiple variants of the same protein, which is a common use case for testing the effects of various single amino acid mutations. We tested PD-var-ILP over increasing coding-sequence lengths and over an increasing number of variants. For the first comparison, we varied the DNA-coding sequence length between 226 and 1626 nucleotides, with increments of 100 nucleotides, using $M = 3$ variants and the same 87-nt upstream and 193-nt downstream sequences as in Appel et al. (2021). For the second comparison, we varied M from 2 to 6 variants. We set $\sigma = 6$ and all other parameter values as in the single-protein case. We compared PD-var-ILP with a greedy baseline, PD-var-Greedy, and evaluated solution efficiency and runtime. We used the same sequence and Linux server as in the single-protein case, and Gurobi for solving the ILP.

When increasing the length of the coding-sequence across three variants, the mean primer efficiency of the optimal PD achieved by PD-var-ILP was higher than or equal to that of PD-var-Greedy in all but one case (Fig. 4A, B; Wilcoxon signed-rank test, $p = 0.0026$), in which PD-var-Greedy obtained a higher mean primer efficiency because it selected fewer primer pairs. PD-var-ILP always achieved an equal or higher total primer efficiency. The mean primer efficiency of both methods generally increased with coding-sequence length, likely due to the increasing number of potential primer pairs in longer sequences. The runtime increased approximately linearly with the coding-sequence length, and the peak memory usage showed a similar linear trend (Supplementary Fig. S4B). When increasing the number of variants, PD-var-ILP found a feasible solution for up to six variants, whereas PD-var-Greedy found a solution for up to four variants only (Fig. 4C, D). For seven or more variants, no feasible solution existed due to the limited number of nonoverlapping primers under the imposed constraints. The mean primer efficiency achieved by PD-var-ILP was always higher than or equal to that obtained by PD-var-Greedy, and its runtime remained under 35 minutes. We also found that the mean primer efficiency of both methods decreased as the number of variants increased, likely due to the additional constraints introduced when more variants must be covered without overlap.

PD-var-Greedy was much faster in both comparisons, which is expected since its runtime is linear in the size of the graph and the number of protein variants desired. However, these runtimes are negligible compared to the time required to perform the experiments, and thus a more efficient PD, which was successfully obtained by PD-var-ILP and is expected to improve PCR amplification efficiency, is desired. Notably, the runtime of both methods remained nearly constant as the number of variants increased, while peak memory usage also remained stable (Supplementary Fig. S4C). For PD-var-ILP, this behavior is expected, as adding variants changes only the flow constraints without increasing the ILP problem size. For PD-var-Greedy, runtime likely stayed stable because each iteration removes overlapping primers, shrinking the graph and speeding up subsequent iterations.

We evaluated the performance of PD-var-ILP on 10 additional proteins, separately. We retrieved the protein-coding sequences from the CCDS database (Pujar et al., 2018) (Supplementary Table S1). We ran PD-var-ILP on each protein with $M = 3$ variants using the aforementioned parameters and compared the mean primer efficiency and runtime to those of PD-var-Greedy. For each protein, PD-var-ILP achieved a mean

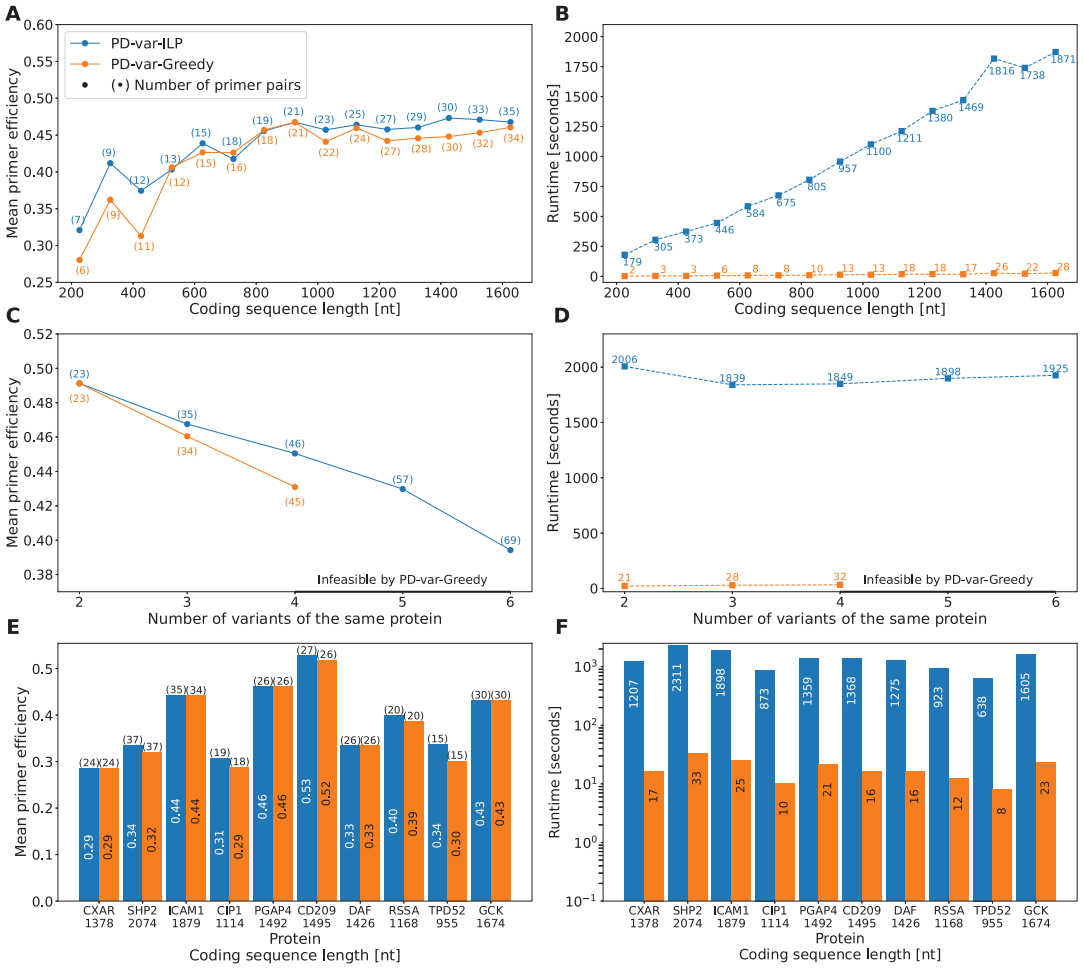


FIG. 4. Performance of PD-var-ILP compared with PD-var-Greedy on multiple variants of the same protein. We gauged the **(A, C, E)** mean primer efficiency and number of primer pairs, and **(B, D, F)** runtime. Numbers in parentheses denote the total number of primer pairs shown for each data point or bar. Protein identifiers are in Supplementary Table S1. ILP, integer linear programming; PD, primer design.

primer efficiency that was greater than or equal to that of PD-var-Greedy (Fig. 4E; $p = 0.0104$, Wilcoxon signed-rank test) and completed in under 40 minutes (Fig. 4F).

5. DISCUSSION

In this study, we formally defined, proved the computational complexity of, and solved the PD problem of selecting primer pairs that guarantee complete coverage of given protein-coding sequences while maximizing primer efficiency and avoiding cross-hybridization. We first proved that the problem is NP-hard, and consequently proposed an ILP formulation for the problem. We then developed algorithms for the different formulations of the problem. PD-single-LPath solves the single-protein case in linear time. For multiple nonhomologous proteins, we introduced both a greedy approach in PD-mul-Greedy and an ILP approach in PD-mul-ILP. For multiple variants of the same protein, we developed a greedy algorithm in PD-var-Greedy and an ILP formulation in PD-var-ILP. We integrated our solutions for the different formulations of the PD problem into a suite of methods, which we call PrimerDesigner.

Our results showed that PrimerDesigner computes optimal PDs which achieve higher efficiency than the solutions obtained by baseline greedy approaches and competing methods, while maintaining reasonable runtime. The higher computational cost of PrimerDesigner reflects its global optimization, whereas competing methods such as PrimalScheme use greedy tiling strategies. Moreover, PrimerDesigner computes optimal solutions

under the specified objective and constraints, thereby providing users with the guarantee that the PD cannot be further improved, unlike competing methods that rely on greedy or heuristic strategies. The computational framework and optimality guarantees of PrimerDesigner, presented in this study, are independent of the specific primer-efficiency function, and thus are applicable to many use cases. Furthermore, we showed that primers selected by PrimerDesigner are predicted to achieve significantly higher PCR efficiency than primers selected by competing methods, demonstrating that optimizing the PrimerDesigner objective translates into meaningful improvements in downstream PCR amplification performance (Supplementary Section S6).

PrimerDesigner provides both exact ILP-based methods and efficient heuristic solutions based on greedy and longest-path algorithms. When forbidden pairs are negligible, such as in single-protein or multiple nonhomologous protein settings (Supplementary Section S2), the longest-path and greedy formulations achieve optimal solutions in practice and are recommended when computational resources are limited. However, in multivariant settings, greedy local decisions may eliminate feasible solutions for other variants. In these cases, the ILP formulation performs global optimization across all variants, achieving higher primer efficiency and identifying feasible solutions for larger variant sets.

While PD is a well-studied problem, the problem solved by PrimerDesigner has unique properties. Existing methods aim to design microarray probes that target as many genes as possible while minimizing cross-hybridization or to design primers for multiplex tiled amplicon sequencing that provide coverage of target sequences while preventing undesired interactions. However, many existing methods rely on heuristic strategies. Optimization-based approaches have been proposed, but they typically address different objectives, such as maximizing coverage or minimizing the number of primers, and do not formulate primer selection as a global optimization problem that maximizes primer efficiency subject to cross-hybridization constraints. In contrast, we focused on designing the most efficient primers that cover complete protein-coding sequences while prohibiting cross-hybridization. While these problems share similarities, both in the NP-hard proof and methodology, they are not equivalent and thus require unique solutions. The PD problem, as we defined and solved it, is critical for high-throughput protein screening, as it enables highly efficient synthesis of libraries of thousands of proteins.

Several questions remain open following our study. First, can we compute the cross-hybridization risk of all possible primer pairs more efficiently? Currently, we compute it using a brute-force algorithm based on the `calc_heterodimer` function. Can we reduce the runtime using efficient sequence-similarity techniques or by modifying the `calc_heterodimer` function to perform pairwise alignment of the whole sequences and use its subalignments for all primer pairs? Second, we solved MEPD-VAR for multiple variants of the same protein using ILP. But is this problem NP-hard or can it be solved in polynomial time? Third, we proved that MEPD-CSP (and consequently MEPD-CMP) are NP-hard with an arbitrary efficiency function. However, given a specific efficiency function that is derived from primer hybridization efficiency—is the problem still NP-hard?

We hope to extend this study in multiple directions in the future. First, we would like to turn the binary restriction of cross-hybridization risk into a more refined risk definition. We plan to maximize efficiency while minimizing the cross-hybridization risk. Moreover, we may also aim for designs with a small number of primer pairs. A solution to these problems will depend on the desired trade-off between primer efficiency, cross-hybridization risk, and the number of primer pairs. This can be achieved using the Lagrangian relaxation, which relaxes constraints by including them in the objective function along with a corresponding penalty (Andreotti et al., 2012). Second, we plan to experimentally test several designs generated by PrimerDesigner and demonstrate their improved protein-synthesis yield compared to the designs generated by the greedy baseline or existing methods. Third, we plan to incorporate degenerate nucleotides into the design to enable more efficient synthesis of protein variants (Shimko et al., 2020). Fourth, we plan to test multiple definitions and parameters of the efficiency function and gauge the performance of PrimerDesigner in terms of efficiency and runtime. Last, we plan to adapt PrimerDesigner to support additional DNA design problems. The underlying formulation can be extended to related settings, such as multiplex tiled amplicon sequencing (Quick et al., 2017), where primers must provide complete coverage while avoiding cross-hybridization, and Golden Gate assembly (Engler et al., 2008), where nodes represent restriction-enzyme cleavage sites, forbidden pairs encode potential misligation events, and edge weights reflect DNA synthesis costs (Hoch et al., 2024). PrimerDesigner's formulations enable globally optimal solutions for a broad class of DNA design problems that are currently addressed using greedy or heuristic methods, with the potential to improve reaction efficiency, reduce experimental failures, and lower design costs.

We expect PrimerDesigner to enable much more efficient synthesis of large protein libraries without cross-hybridization. Such synthesis will further advance protein research using high-throughput screening techniques and, consequently, deliver more accurate machine-learning protein-function models and an ability to design proteins with improved function for therapeutics and other applications.

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AUTHORS' CONTRIBUTIONS

J.M.: Methodology, software, validation, visualization, investigation, writing—original draft, and formal analysis. M.B.: Investigation, methodology, and formal analysis. S.L.: Methodology and investigation. P.F.: Conceptualization, funding acquisition, and supervision. Y.O.: Conceptualization, supervision, project administration, and writing—review and editing.

DATA AVAILABILITY STATEMENT

PrimerDesigner is available at: https://github.com/OrensteinLab/PrimerDesigner. The data and sequences used in this study are available in the GitHub repository.

AUTHOR DISCLOSURE STATEMENT

No competing financial interests exist.

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SUPPLEMENTARY MATERIAL

Supplementary Data

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